

CFO Financial Command Center — Apex Distribution Ltd.

Power BI Executive Dashboard Case Study

1. Executive Summary

I designed and built the CFO Financial Command Center for Apex Distribution Ltd., a fictional mid-sized distribution company. This portfolio simulation demonstrates an end-to-end executive financial reporting and scenario-planning solution built entirely in Power BI Desktop. The dataset is synthetic and was created for demonstration purposes only. No real client was involved, and no actual business outcomes, ROI, or testimonials are claimed.

The solution answers ten critical questions that a Chief Financial Officer asks during monthly reviews: Is the company growing? Is growth translating into profit? Is gross margin improving? Where are costs increasing? Which regions and customer segments are most profitable? Is the business overly dependent on a few customers? Which products are below the required margin? Is actual performance meeting budget? Is the company generating enough cash? And how would revenue growth, COGS reductions, or OPEX reductions affect operating profit?

The final deliverable consists of three integrated dashboard pages — Executive Overview, Customer & Product Performance, and CFO Review & Scenario Planning — supported by a star-schema data model, a dedicated fiscal calendar, time-intelligence DAX measures, dynamic KPI cards with sparklines, RAG conditional formatting, customer concentration analysis, product profitability scatter plots, budget variance matrices, cash-flow and runway analysis, and interactive what-if scenario planning.

2. Business Context

Apex Distribution Ltd. is a fictional mid-sized distribution company operating across multiple European regions and serving a diversified customer base spanning enterprise, mid-market, and small-business segments. The company distributes products across categories including electronics, furniture, industrial supplies, and office equipment.

In this simulated environment, the CFO was responsible for consolidating fragmented financial data into a single analytical system that could support monthly board reviews, investor updates, and operational decision-making. The existing reporting was assumed to be spreadsheet-based, manually compiled, and too slow to support scenario planning or real-time variance analysis.

3. Stakeholder Questions

The primary stakeholder — the Chief Financial Officer — needed a consolidated system to answer the following questions:

- Is the company growing, and at what rate?
- Is top-line growth translating into bottom-line profit?
- Is gross margin improving or deteriorating over time?
- Where are costs increasing fastest, and is that growth justified?
- Which regions and customer segments are the most and least profitable?
- Is the company overly dependent on a small number of customers?
- Which products are performing below the required gross margin threshold?
- Is actual financial performance meeting the approved budget?
- Is the company generating sufficient cash, and what is the liquidity runway?
- How would specific operational levers — revenue growth, COGS efficiency, or OPEX reduction — impact operating profit under alternative assumptions?

4. Project Objectives

My objectives for this portfolio project were:

- Build a production-quality star-schema model with a dedicated fiscal calendar to support time-intelligence analysis.
- Create dynamic KPI cards that display the current value, prior-year comparison, percentage change, and sparkline trend in a single visual unit.
- Implement metric-aware RAG (Red-Amber-Green) logic that treats revenue and profit growth as favorable, while treating increases in COGS, discounts, and operating expenses as unfavorable.
- Develop customer concentration analysis to quantify revenue dependency risk.
- Build a product profitability portfolio scatter plot to identify underperforming products against a 32% gross margin target.
- Deliver budget variance analysis comparing actuals to budget across revenue, gross profit, and operating profit.
- Calculate net cash flow, operating cash flow, and cash runway to assess liquidity health.
- Implement interactive what-if parameters for revenue growth, COGS reduction, and OPEX reduction to enable scenario planning.
- Ensure all dynamic titles, subtitles, labels, and color codes respond to user selections and scenario assumptions.

5. Dataset and Data Quality

The dataset is entirely synthetic and was generated for the purpose of this portfolio simulation. Apex Distribution Ltd. is a fictional entity, and all financial figures, customer names, product names, and regional data are fabricated. I designed the dataset to reflect a realistic mid-sized distribution business with multiple dimensions: time, region, customer segment, product category, and individual products.

The raw data included transactional revenue records, product cost data, operating expense line items, budget targets, and cash-flow movements. From the outset, I documented the synthetic nature of the data transparently. I did not attempt to hide data limitations or present the figures as empirical truth. This transparency is essential for portfolio integrity and recruiter trust.

6. Power Query Transformation Process

I used Power Query in Power BI Desktop to ingest, clean, and transform the raw synthetic data. Several transformation steps were critical to producing a coherent management-reporting dataset.

First, the original AI-generated expense data produced unrealistic financial ratios. Some expense values were disproportionately large relative to revenue, which would have created a nonsensical operating profit scenario. I normalized expense values in Power Query to create a coherent management-reporting scenario where the income statement, balance sheet, and cash-flow figures remained internally consistent.

Second, the revenue date field was originally imported as text. This caused fiscal-year slicers and time-intelligence DAX measures to return blank values. I diagnosed the issue by inspecting the data type in Power Query, converting the text field to a proper date type, and verifying that the resulting date range covered the full fiscal periods required for year-over-year comparisons.

Third, I removed duplicates, standardized category names, and created surrogate keys for the dimension tables to support the star-schema relationships in the data model.

7. Data Model and Star Schema

I modeled the data using a classic star schema. The fact table contains transactional financial data at the grain of individual sales transactions, linked to dimension tables for date, region, customer, product, and segment.

A dedicated fiscal calendar dimension table was essential. Standard calendar tables do not align with fiscal reporting periods, which can break year-over-year and year-to-date calculations. I built a custom date table with fiscal year, fiscal quarter, and fiscal period columns, and marked it as the official date table in Power BI. This ensured that all time-intelligence measures — SAMEPERIODLASTYEAR, TOTALYTD, and custom YoY formulas — returned accurate results.

The relationships were configured as one-to-many from each dimension table to the fact table, with single-direction cross-filtering to preserve performance and prevent ambiguous aggregation paths. I also created a separate budget fact table at a higher grain and used DAX to bridge actuals and budget for variance analysis.

8. DAX Architecture

The DAX layer is the analytical engine of this solution. I organized measures into logical groups: revenue, cost, profit, margin, growth, budget variance, cash flow, and scenario planning.

For time-intelligence, I used a combination of built-in functions and custom logic. Year-over-year growth measures compare the current fiscal period to the same period in the prior fiscal year using `SAMEPERIODLASTYEAR`. I also created dynamic prior-year labels that update based on the selected fiscal year, so KPI cards always display the correct comparison period.

For the KPI cards, I built hybrid visuals because a single native Power BI card could not reproduce the required layout — large current value, smaller prior-year label, percentage change with directional arrow, and a sparkline trend chart. I combined shapes, card visuals, dynamic text measures, and sparkline line charts to achieve the desired executive look.

Percentage-change calculations involving negative operating profit initially produced misleading directional indicators. When the prior-year operating profit is negative, a standard percentage-change formula can invert the business meaning. I corrected the operating-profit trend formula by using the absolute value of the prior-year profit as the denominator, ensuring that the directional arrow and color coding accurately reflect whether performance improved or deteriorated.

9. Financial Business Logic

The financial logic in this dashboard goes beyond simple aggregation. I implemented metric-aware RAG conditional formatting that understands the business direction of each metric.

For revenue, gross profit, and operating profit, a positive year-over-year change is favorable and is colored green. For COGS, operating expenses, and discounts, a positive year-over-year change is unfavorable and is colored red or amber. This required reversing the status logic for cost metrics. Higher discounts, COGS, and OPEX are unfavorable even when their mathematical YoY change is positive. I encoded this logic explicitly in DAX to prevent misinterpretation by executives.

For the financial performance breakdown table, I calculated each metric as a percentage of net revenue to give the CFO immediate context on cost structure and profitability composition.

10. Dashboard UX and Design System

I designed the dashboard with a dark executive theme to reduce eye strain during boardroom presentations and to create visual hierarchy. The color system uses green for favorable movements, red for unfavorable movements, and amber for watch-list items. All text is sized for readability at a distance, with the largest font reserved for the primary KPI value, medium font for the comparison label, and small font for supporting context.

Navigation is handled through a persistent top tab bar allowing the CFO to switch between Executive Overview, Customer & Product, and CFO Review pages without losing filter context. Slicers for fiscal year, fiscal quarter, region, segment, and category are positioned consistently across pages to maintain a predictable user experience.

11. Executive Overview Walkthrough

The Executive Overview page is the landing view for the CFO and board members. It provides a single-screen summary of financial health.

Four executive KPI cards display the headline figures: Net Revenue of \$816.16K with +126.0% year-over-year growth; Gross Profit of \$238.06K with +117.7% growth; Operating Profit of \$98.07K with +144.3% growth; and Gross Margin of 29.17%, which is down 1.1 percentage points from the prior year. Each card includes a sparkline showing the intra-year trend and a dynamic comparison label.

Below the KPIs, a profitability heatmap shows gross margin percentage by region and customer segment. This immediately reveals geographic and segment-level performance patterns. For example, some regions show healthy margins above 32% while others fall into the critical or watch zones.

The financial performance breakdown table lists net revenue, COGS, gross profit, operating expenses, operating profit, and discounts with current values, percentages of net revenue, YoY growth rates, and RAG performance indicators. The narrative this page tells is clear: the company generated strong revenue and profit growth, but gross margin declined. Cost growth therefore requires attention even though absolute profit increased.

12. Customer & Product Performance Walkthrough

The Customer & Product Performance page shifts the lens from aggregate financials to operational drivers.

The KPI cards show: 80 Active Customers with +128.6% growth; Revenue per Customer of \$10.2K, down 1.1% year-over-year; Top-Five Customer Concentration of 47.9%, with the top five customers generating approximately \$391.3K in revenue; and 12 At-Risk Products out of 32 active products, representing 37.5% of the portfolio operating below the 32% gross margin target.

The customer revenue concentration chart displays individual customer bars sorted by revenue, overlaid with a cumulative percentage line. This Pareto-style visualization makes concentration risk immediately visible. The product profitability portfolio scatter plot places each product on axes of net revenue (X) and gross margin (Y), with bubble size representing sales volume. A 32% margin target line and an average revenue reference line divide the plot into quadrants, allowing the CFO to identify star products, cash cows, and underperformers at a glance.

The narrative here is that the company expanded its customer base significantly, but revenue per customer declined slightly. Almost half of total revenue is generated by the five largest customers, creating concentration risk. A significant portion of the product portfolio also operates below the target margin.

13. CFO Review & Scenario Planning Walkthrough

The CFO Review & Scenario Planning page combines budget variance analysis, cash-flow assessment, and interactive scenario modeling.

The KPI cards show: Actual Revenue of \$816.16K against a budget of \$840.6K, representing a -2.9% variance; Actual Operating Profit of \$98.07K against a budget of \$108.4K, a -9.5% variance; Net Cash Flow of \$2.53M with Operating Cash Flow of approximately \$1.9M; and Cash Runway of 42.9 months against a target of at least 6 months.

The budget versus actual performance matrix compares actual values, budget values, absolute variance, variance percentage, and status for net revenue, gross profit, and operating profit. This gives the CFO a structured view of where the business is ahead or behind plan.

The scenario planning section uses three what-if parameters: Revenue Growth at +21%, COGS Reduction at 7%, and OPEX Reduction at 5%. These sliders feed into DAX measures that calculate the projected operating profit under the combined assumptions. The displayed scenario result shows a projected operating profit of approximately \$204.0K, an improvement of approximately +108% versus current operating profit. The contribution breakdown reveals that revenue growth contributes the largest dollar impact, followed by COGS efficiency, then OPEX savings.

The narrative is that the company is below budget on revenue and operating profit but maintains strong liquidity. Scenario analysis indicates that revenue growth and COGS efficiency are the most significant levers for improving operating profit.

14. Key Business Insights

From the analysis, the following insights emerge:

- Revenue and profit are growing rapidly, but gross margin compression indicates that cost management is not keeping pace with scale.

- Customer acquisition is strong, yet revenue per customer has declined slightly, suggesting that newer customers may be smaller or more price-sensitive.
- Customer concentration is elevated: the top five customers account for nearly half of all revenue, creating vulnerability if any major customer churns.
- Over one-third of the product portfolio operates below the 32% margin target, indicating a need for pricing review, cost reduction, or product rationalization.
- The company is below budget on both revenue and operating profit, requiring management attention to close the gap in the remaining fiscal periods.
- Liquidity is healthy, with a cash runway of 42.9 months, providing operational flexibility to invest in margin-improvement initiatives.
- Scenario modeling shows that a combination of revenue growth and COGS efficiency could more than double operating profit, giving the CFO a quantified case for operational investment.

15. Recommendations for Management

Based on the dashboard insights, I would recommend the following actions:

- Investigate gross margin deterioration by region and segment to identify whether the issue is driven by pricing, product mix, or logistics costs.
- Develop a customer diversification strategy to reduce dependency on the top five accounts, including targeted prospecting in underpenetrated segments.
- Conduct a product portfolio review for the 12 at-risk products to determine whether margins can be improved through supplier renegotiation, price increases, or discontinuation.
- Address the revenue and operating profit budget shortfalls by diagnosing whether the miss is due to market conditions, execution gaps, or overly optimistic budget assumptions.
- Use the scenario planning model in monthly CFO reviews to stress-test assumptions and align the leadership team around quantified improvement targets.

16. Validation and Quality Assurance

I validated the dashboard through multiple quality-assurance steps. First, I reconciled all summary-level KPIs against the underlying transactional data to ensure that filters and relationships were aggregating correctly. Second, I tested time-intelligence measures across different fiscal year and quarter selections to confirm that prior-year comparisons shifted correctly.

Third, I verified the budget variance calculations by independently computing the expected variance percentages and confirming they matched the DAX output. Fourth, I tested the scenario planning sliders across their full ranges to ensure that the projected operating profit updated smoothly and that the contribution breakdowns summed correctly to the total projected profit.

Finally, I reviewed the RAG logic manually for each metric to confirm that favorable and unfavorable directions were encoded correctly, particularly for cost metrics where the business logic is reversed.

17. Development Challenges and Resolutions

I encountered and resolved several technical challenges during development:

- Unrealistic expense ratios in the original synthetic data required normalization in Power Query to create a coherent management-reporting scenario.
- The revenue date field imported as text, breaking fiscal-year slicers and time-intelligence measures. I corrected this by converting the field to a date type and validating calendar coverage.
- KPI cards required a hybrid design using shapes, card visuals, dynamic labels, and sparklines because one native Power BI card could not reproduce the required executive layout.
- Percentage-change calculations involving negative operating profit produced misleading directions. I corrected the formula by using the absolute value of prior-year profit as the denominator.
- Status logic had to be reversed for cost metrics. Higher discounts, COGS, and OPEX are unfavorable even when their mathematical YoY change is positive. I encoded this explicitly in DAX.
- A native waterfall visual was unavailable for the financial bridge, so I created a custom financial bridge matrix using a combination of DAX measures and conditional formatting to show the step-by-step contribution of each driver.
- Synthetic data limitations were documented transparently rather than hidden, maintaining portfolio integrity.

18. Skills Demonstrated

This project demonstrates the following technical and analytical competencies:

- Power BI Desktop — report design, visual composition, and interactivity.
- Power Query — data ingestion, cleaning, normalization, and transformation.
- DAX — time-intelligence measures, dynamic labels, conditional logic, and scenario modeling.
- Star-schema data modeling — dimension and fact table design, relationship management, and fiscal calendar implementation.
- Financial analytics — income statement logic, margin analysis, budget variance, cash-flow modeling, and runway calculation.
- UX design for executives — dark theme, visual hierarchy, RAG color coding, and page navigation.
- Business communication — translating raw data into structured insights that answer what happened, why it happened, where to act, and what could happen.

19. Limitations

This is a portfolio simulation. Apex Distribution Ltd. is fictional, and all data is synthetic. No real business outcomes, ROI figures, adoption rates, or client testimonials are claimed. The dataset was designed to demonstrate analytical techniques rather than to reflect a live operational environment. Some real-world complexities — such as multi-currency consolidation, intercompany eliminations, and detailed general-ledger mappings — are simplified or omitted to keep the project focused on core dashboard and DAX skills.

20. Future Enhancements

If I were to extend this project, I would consider the following enhancements:

- Integrate actual general-ledger data with automated refresh via Power BI Service and scheduled dataflows.
- Add drill-through pages for individual customers and products to support operational deep-dives.
- Implement row-level security (RLS) so that regional managers see only their territory while the CFO sees everything.
- Add forecasting visuals using time-series decomposition or integration with Python/R forecasting libraries.
- Build a mobile-optimized layout for executive consumption on tablets and phones.
- Expand the scenario model to include sensitivity analysis and tornado charts showing which assumptions have the greatest impact on operating profit.

21. Recruiter Takeaway

This case study demonstrates my ability to take raw synthetic data, apply rigorous data-quality discipline, model it into a star schema, build sophisticated DAX measures, and deliver a visually polished, analytically rigorous executive dashboard that answers real CFO questions. I do not claim this was a live client deployment, and I am transparent about the synthetic nature of the data. What I do demonstrate is end-to-end ownership of the BI development lifecycle — from data preparation through modeling, visualization, and business insight — with a clear focus on executive decision support.

22. Conclusion

The CFO Financial Command Center for Apex Distribution Ltd. is a comprehensive portfolio simulation that showcases advanced Power BI development skills applied to a realistic financial analytics scenario. By combining star-schema modeling, time-intelligence DAX, metric-aware RAG logic, customer concentration analysis, product profitability assessment, budget variance tracking, cash-flow analysis, and interactive scenario planning, I built a three-page executive decision-support system that tells a coherent financial story.

The dashboard answers what happened through historical KPIs and trends, why it happened through margin and concentration analysis, where management should act through RAG indicators and at-risk product flags, and what could happen under alternative assumptions through dynamic scenario modeling. For recruiters, hiring managers, and prospective clients, this project represents a transparent, technically sound, and commercially aware demonstration of my capabilities as a Business Intelligence consultant.